

LAB EXPERIMENT RECORD

Set 4: Product Inventory Management

Department	AIML
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	S.Naveen Kumar
Register Number	192425176
Date	1 October 2026
Day	Thursday

Aim

To visualize product inventory data using a bar chart, stacked bar chart, and scatter plot in R, and to build an interactive dashboard in Tableau.

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Supplementary Data: Category and Price (assumed)

Product Name	Category	Price (\$)
Product A	Electronics	50
Product B	Electronics	80
Product C	Home	30
Product D	Home	60
Product E	Office	40

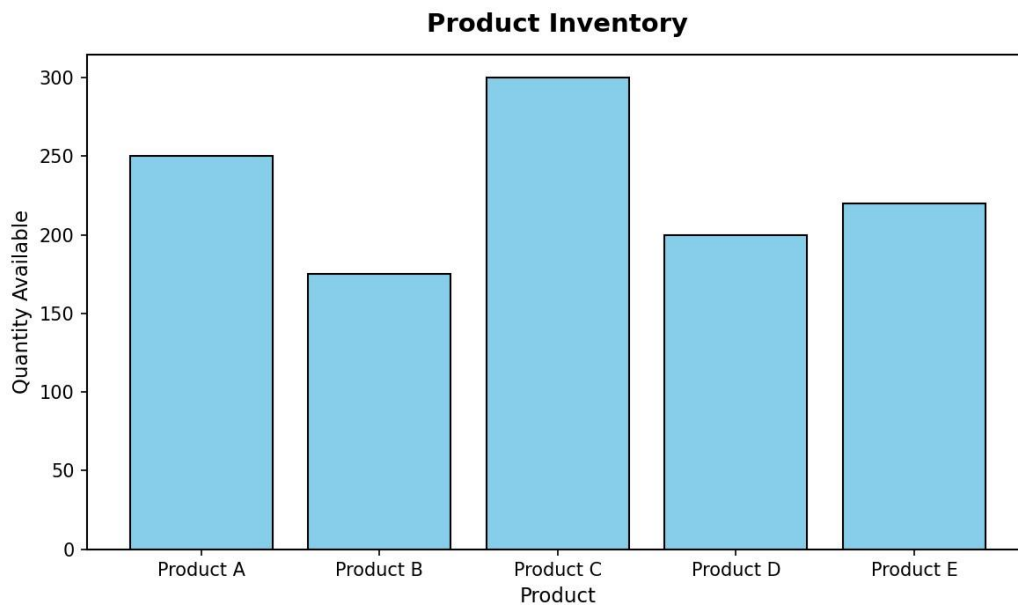
Note: The assignment table contains only Product ID, Product Name, and Quantity Available. The Category and Price values above are assumed sample values used for Programs 2 and 3.

PROGRAM 1: Bar Chart of Product Quantity

Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Dataset product <- c("Product A", "Product B", "Product C", "Product D",
"Product E") quantity <- c(250, 175, 300, 200, 220)
```

```
# Bar Chart barplot(quantity,
names.arg = product,      xlab =
"Product",                ylab = "Quantity
Available",               main = "Product
Inventory",               col = "skyblue")
```



Output 1: Product Inventory Bar Chart

Observation: Product C has the highest quantity (300 units) and Product B has the lowest (175 units). Products A, D, and E lie in between at 250, 200, and 220 units.

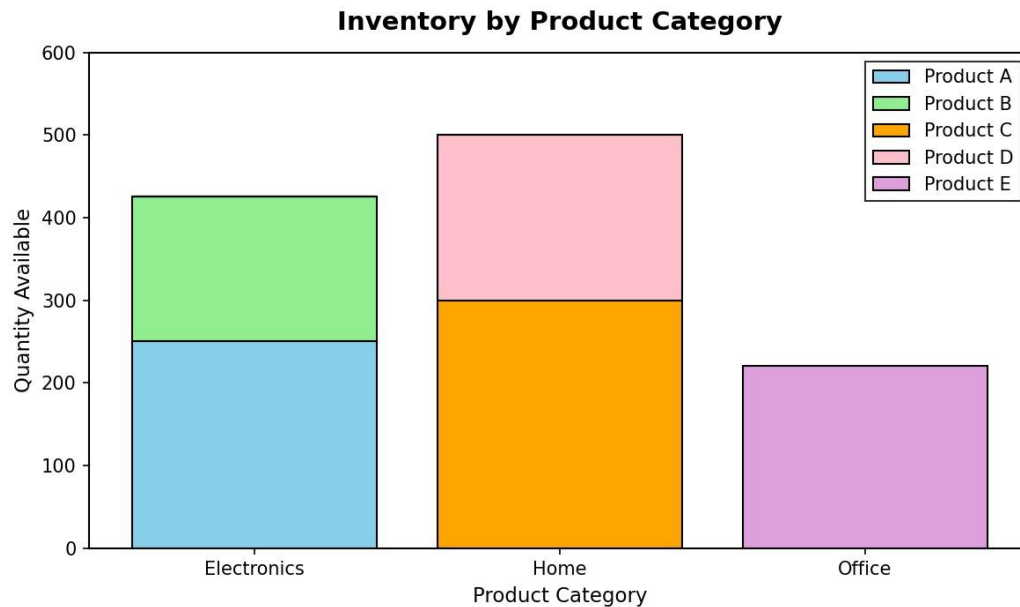
PROGRAM 2: Stacked Bar Chart by Product Category

Generate a stacked bar chart to show the quantity of each product within different product categories.

```
# Quantity of each product within each category
inventory <- matrix(c(250, 175, 0, 0, 0,
0, 0, 300, 200, 0,
0, 0, 0, 0, 220),      nrow = 5,
dimnames = list(      c("Product A", "Product B", "Product C", "Product
D", "Product E"),      c("Electronics", "Home", "Office")))

# Stacked Bar Chart barplot(inventory,      beside = FALSE,
xlab = "Product Category",      ylab = "Quantity Available",
main = "Inventory by Product Category",      col = c("skyblue",
"lightgreen", "orange", "pink", "plum"),      ylim = c(0, 600),
```

```
legend.text = rownames(inventory),      args.legend = list(x =
"topright"))
```



Output 2: Inventory by Product Category (Stacked Bar Chart)

Observation: The Home category holds the most stock (500 units: Product C 300 and Product D 200), followed by Electronics (425 units: Product A 250 and Product B 175). Office has 220 units, all from Product E.

PROGRAM 3: Scatter Plot of Price vs Quantity Available

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
# Dataset price <- c(50, 80, 30, 60,
40) quantity <- c(250, 175, 300, 200,
220)

# Scatter Plot plot(price, quantity,      main =
"Product Price vs Quantity Available",      xlab =
"Product Price ($)",      ylab = "Quantity
Available",      pch = 19,      col = "red")

# Correlation coefficient
cor(price, quantity)
```



Output 3: Product Price vs Quantity Available Scatter Plot

Insight: The points slope downward: as the product price increases, the quantity available decreases. The correlation coefficient is about -0.87, a strong negative relationship. For example, Product C (\$30) has 300 units in stock while Product B (\$80) has only 175 units. Cheaper products are stocked in larger quantities, which may reflect higher demand for lower-priced items. Since the price values are assumed and only five products are used, this is an indication rather than a firm conclusion.

PROGRAM 4: Tableau Dashboard for Product Inventory

Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Steps in Tableau:

1. Save the inventory data as an Excel or CSV file with the columns Product ID, Product Name, Category, Price, and Quantity Available.
2. Open Tableau, choose Connect > Microsoft Excel / Text file, and load the file. Open Sheet 1.
3. Sheet 1 (Bar Chart): drag Product Name to Columns and Quantity Available to Rows (SUM). Choose the Bar marktype and title the sheet "Product Inventory".
4. Sheet 2 (Stacked Bar Chart): drag Category to Columns, Quantity Available to Rows (SUM), and Product Name to Color. This stacks each product inside its category. Title the sheet "Inventory by Category".
5. Create a new Dashboard, drag Sheet 1 and Sheet 2 onto it, and arrange them side by side.
6. For interactivity, select "Use as Filter" on both charts and add a Category filter (Show Filter), so clicking a bar updates the other chart.
7. Title the dashboard "Product Inventory Dashboard" and save the workbook.

Insert Tableau dashboard screenshot here (Output 4)

Output 4: Product Inventory Tableau Dashboard

Observation: The dashboard shows total quantity per product and the category-wise stacking together. Selecting a bar or a category filters both charts, so users can explore the inventory interactively.

Set 4 Procedure

1. Create vectors for the product names, quantities, and prices.
2. Use `barplot()` to display the quantity of each product.
3. Create a product-by-category matrix and use `barplot()` with `beside = FALSE` for the stacked bar chart.
4. Use `plot()` for the scatter plot of price against quantity and `cor()` to measure the relationship.
5. Connect the dataset in Tableau, build the two sheets, and combine them in a dashboard with filters.

Set 4 Result

The product inventory data was successfully visualized using a bar chart, stacked bar chart, and scatter plot in R. A Tableau dashboard combining the bar chart and stacked bar chart was created for interactive exploration.

Key Functions Used

Function / Tool	Purpose
barplot()	Creates the bar chart and stacked bar chart.
matrix()	Builds the product-by-category quantity table.
plot()	Creates the scatter plot.
cor()	Calculates the correlation coefficient.
Tableau Dashboard	Combines sheets with filters for interactive exploration.

SET 5: WEBSITE ANALYTICS

Dataset: Website Traffic

Aim

To analyse website traffic data using a line chart, bar chart, and stacked area chart in R, and to build an interactive dashboard in Tableau.

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

Supplementary Data: User Interactions (assumed)

Date	Likes	Shares	Comments
2023-01-01	300	120	60
2023-01-02	340	140	70
2023-01-03	280	100	50
2023-01-04	360	150	80
2023-01-05	400	170	90

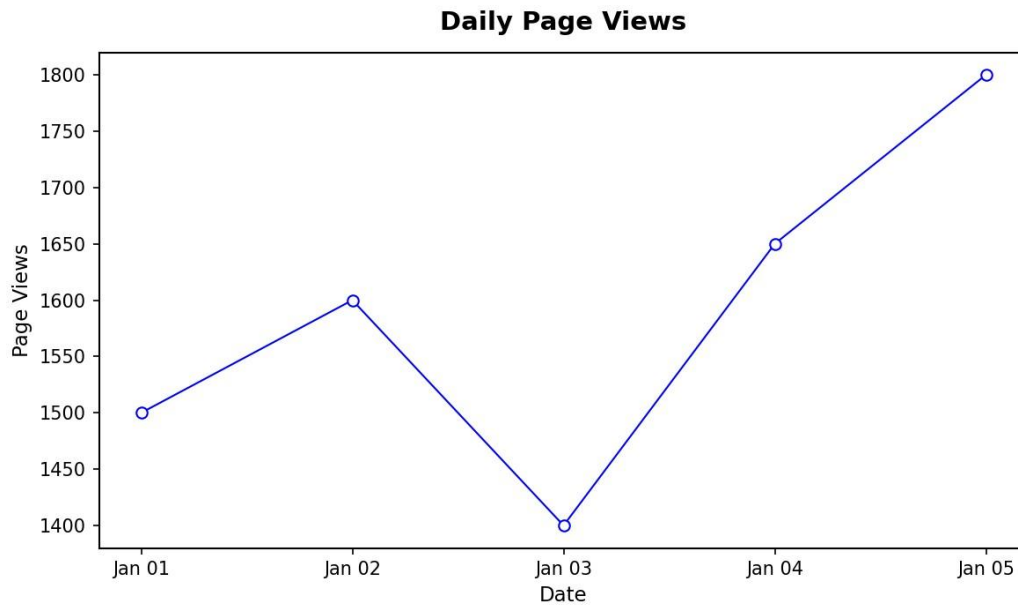
Note: The assignment table contains only Date, Page Views, and Click-through Rate. The likes, shares, and comments values above are assumed sample values used for Program 3.

PROGRAM 1: Line Chart of Daily Page Views

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Dataset date <- as.Date(c("2023-01-01", "2023-01-02",
"2023-01-03", "2023-01-04", "2023-01-05"))
page_views <- c(1500, 1600, 1400, 1650, 1800)

# Line Chart plot(date,
page_views,      type = "o",
xlab = "Date",   ylab =
"Page Views",    main =
"Daily Page Views", col =
"blue")
```



Output 1: Daily Page Views Line Chart

Observation: Page views rise from 1500 on January 1 to 1600 on January 2, fall to the lowest value of 1400 on January 3, and then increase to 1650 and finally 1800, the highest value, on January 5.

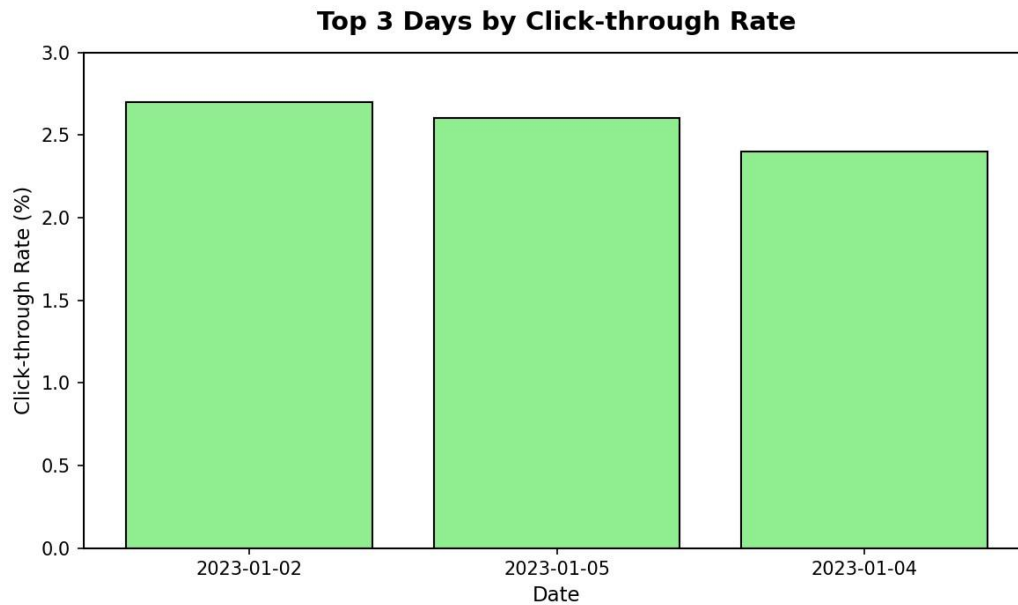
PROGRAM 2: Bar Chart of Top N Days by Click-through Rate

Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Dataset date <- as.Date(c("2023-01-01", "2023-01-02",
  "2023-01-03", "2023-01-04", "2023-01-05")) ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6) traffic <-
data.frame(date, ctr)

# Select the top N days
N <- 3
top_days <- head(traffic[order(-traffic$ctr), ], N)

# Bar Chart barplot(top_days$ctr, names.arg =
format(top_days$date, "%Y-%m-%d"), xlab =
"Date", ylab = "Click-through Rate (%)",
main = "Top 3 Days by Click-through Rate", col
= "lightgreen", ylim = c(0, 3))
```



Output 2: Top 3 Days by Click-through Rate

Observation: With $N = 3$, the best days are 2023-01-02 (2.7%), 2023-01-05 (2.6%), and 2023-01-04 (2.4%). January 2 has the highest click-through rate, while January 3 (2.0%) is the lowest overall.

PROGRAM 3: Stacked Area Chart of User Interactions

Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

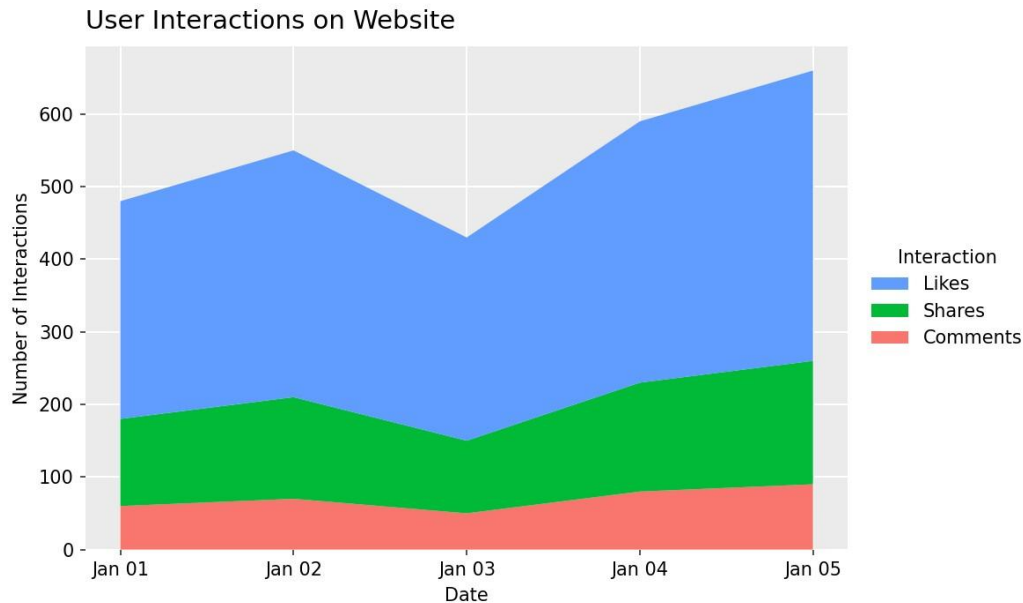
```
library(ggplot2)

# Dataset (assumed interaction counts) date <-
as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
"2023-01-04", "2023-01-05"))

interactions <- data.frame(
  Date = rep(date, times = 3),
  Type = rep(c("Likes", "Shares", "Comments"), each = 5),
  Count = c(300, 340, 280, 360, 400,
            120, 140, 100, 150, 170,
            60, 70, 50, 80, 90))

interactions$Type <- factor(interactions$Type,
                           levels = c("Likes", "Shares", "Comments"))

# Stacked Area Chart ggplot(interactions, aes(x = Date, y =
Count, fill = Type)) + geom_area() + labs(title = "User
Interactions on Website", x = "Date", y =
"Number of Interactions", fill = "Interaction")
```

Output 3: User Interactions Stacked Area Chart

Observation: Likes form the largest share of interactions every day, followed by shares and then comments. Total interactions dip on January 3 and reach their peak on January 5, following the same pattern as page views.

PROGRAM 4: Tableau Dashboard for Website Traffic

Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Steps in Tableau:

1. Save the website traffic data as an Excel or CSV file with the columns Date, Page Views, and Click-through Rate(store the rate as a number, for example 2.3).
2. Open Tableau, connect to the file, and make sure Date is set to the Date data type.
3. Sheet 1 (Line Chart): drag Date to Columns (Day, continuous) and Page Views to Rows (SUM). Choose the Linemark type and title the sheet "Daily Page Views".
4. Sheet 2 (Bar Chart): drag Date to Columns and Click-through Rate to Rows (AVG). Sort the bars in descending order. To show only the top N days, right-click Date on the Filters shelf > Top > By field > Top 3 by Click-through Rate. Title the sheet "Top Days by Click-through Rate".
5. Create a new Dashboard, drag Sheet 1 and Sheet 2 onto it, and arrange them one above the other.
6. Select "Use as Filter" on the bar chart and add a Date range filter so users can explore different periods.
7. Title the dashboard "Website Traffic Dashboard" and save the workbook.

Insert Tableau dashboard screenshot here (Output 4)

Output 4: Website Traffic Tableau Dashboard

Observation: The dashboard shows the page-view trend and the best click-through days together. Selecting a day in the bar chart or changing the date filter updates the view for interactive exploration.

Set 5 Procedure

1. Create vectors for the dates, page views, and click-through rates.
2. Use `plot()` with `type = "o"` to draw the line chart of daily page views.
3. Sort the data with `order()`, select the top N rows with `head()`, and use `barplot()` to draw the bar chart.
4. Create a long-format data frame of likes, shares, and comments, and use `ggplot()` with `geom_area()` for the stacked area chart.
5. Connect the dataset in Tableau, build the line chart and bar chart sheets, and combine them in a dashboard.

Set 5 Result

The website traffic data was successfully visualized using a line chart, bar chart, and stacked area chart in R. A Tableau dashboard combining the line chart and bar chart was created for interactive exploration.

Key Functions Used

Function / Tool	Purpose
<code>as.Date()</code>	Converts text into Date values.
<code>plot()</code>	Creates the line chart.
<code>order()</code> / <code>head()</code>	Sorts the data and selects the top N rows.
<code>barplot()</code>	Creates the bar chart.
<code>ggplot()</code> + <code>geom_area()</code>	Creates the stacked area chart.
Tableau Dashboard	Combines sheets with filters for interactive exploration.

SET 6: PRODUCT SALES ANALYSIS

Dataset: Monthly Product Sales

Aim

To analyse monthly product sales using a grouped bar chart, stacked area chart, and data table in R, and to build an interactive dashboard in Tableau.

Dataset: Monthly Product Sales

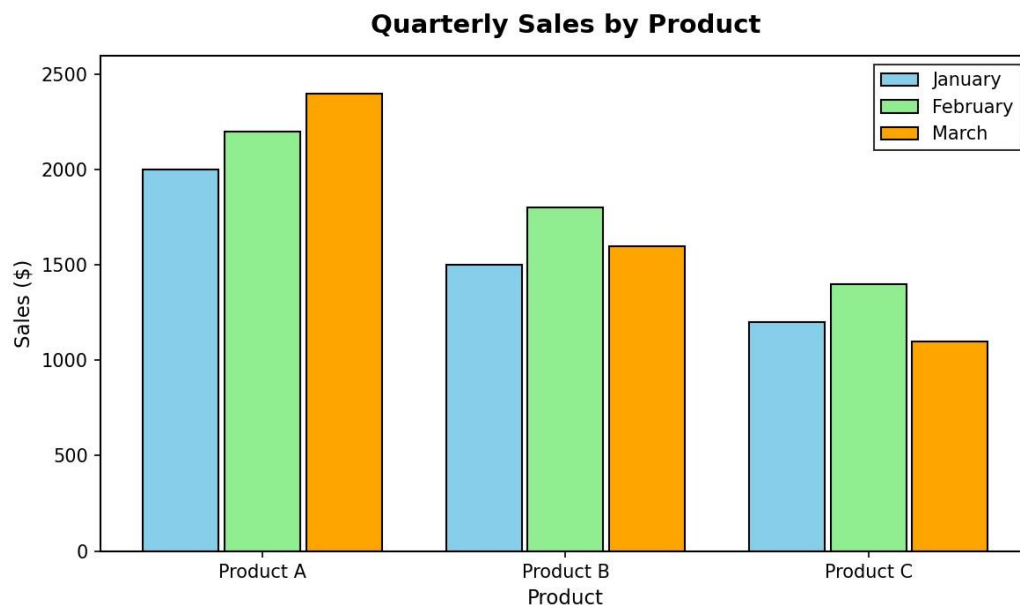
Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

PROGRAM 1: Grouped Bar Chart of Quarterly Sales

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
# Dataset (rows = months, columns = products)
sales <- matrix(c(2000, 2200, 2400,
                  1500, 1800, 1600,
                  1200, 1400, 1100),
               nrow = 3,
               dimnames = list(
                 c("January", "February", "March"),
                 c("Product A", "Product B", "Product C")))

# Grouped Bar Chart barplot(sales, beside =
TRUE, xlab = "Product", ylab =
"Sales ($)", main = "Quarterly Sales by
Product", col = c("skyblue", "lightgreen",
"orange"), ylim = c(0, 2600),
legend.text = rownames(sales), args.legend
= list(x = "topright"))
```



Output 1: Quarterly Sales by Product (Grouped Bar Chart)

Observation: Product A has the highest sales in every month and grows steadily from 2000 to 2400. Product B peaks in February (1800) and Product C also peaks in February (1400) before falling in March.

PROGRAM 2: Stacked Area Chart of Overall Sales Trend

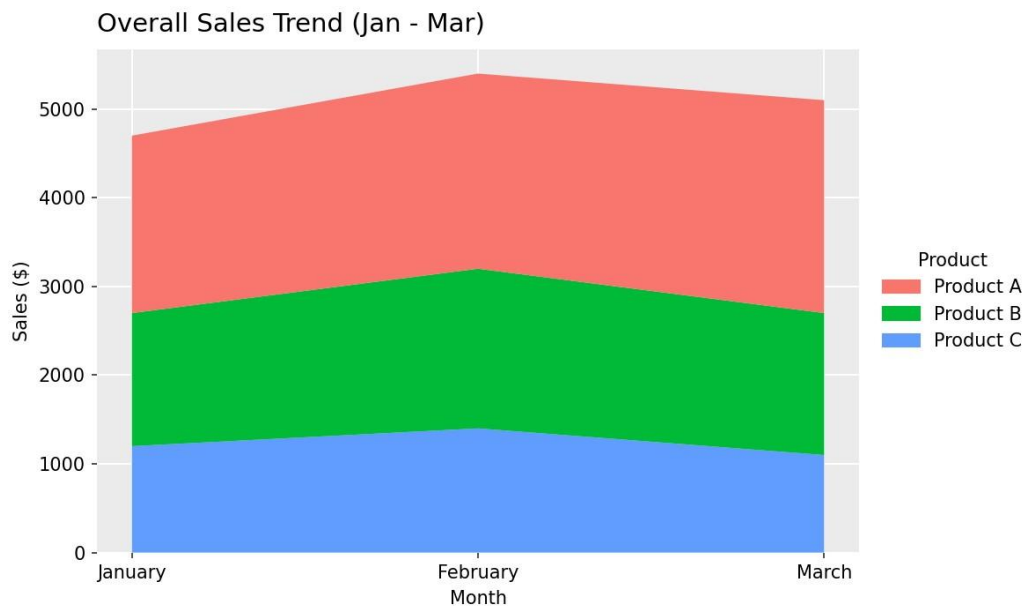
Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
library(ggplot2)

# Long-format dataset sales_long <- data.frame(  Month =
rep(c("January", "February", "March"), times = 3),
  Product = rep(c("Product A", "Product B", "Product C"), each = 3),
  Sales = c(2000, 2200, 2400,
            1500, 1800, 1600,
            1200, 1400, 1100))

sales_long$Month <- factor(sales_long$Month,
                           levels = c("January", "February", "March"))

# Stacked Area Chart ggplot(sales_long, aes(x = Month, y = Sales, fill =
Product, group = Product)) + geom_area() + labs(title = "Overall Sales Trend
(Jan - Mar)", x = "Month", y = "Sales ($)")
```



Output 2: Overall Sales Trend (Stacked Area Chart)

Observation: Combined sales of all three products are 4700 in January, rise to 5400 in February, and dip to 5100 in March. Product A forms the largest band throughout.

PROGRAM 3: Table of Monthly Sales Data

Build a table to show the monthly sales data for each product. Label the table elements.

```
# Dataset sales_data <-
data.frame(  Product_ID =
c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100))

# Total sales for the quarter
sales_data$Total_Sales <- rowSums(sales_data[, 3:5])

# Display the table
print(sales_data)
```

Product ID	Product Name	January Sales	February Sales	March Sales	Total Sales
1	Product A	2000	2200	2400	6600
2	Product B	1500	1800	1600	4900
3	Product C	1200	1400	1100	3700

Output 3: Monthly Product Sales Table

Observation: The table lists each product's sales for January, February, and March with the quarterly total. Product A has the highest total (6600), followed by Product B (4900) and Product C (3700).

PROGRAM 4: Tableau Dashboard for Product Sales

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Steps in Tableau:

1. Save the sales data as an Excel or CSV file. In Tableau, select the January, February, and March Sales columns, right-click, and choose Pivot to create the fields Pivot Field Names (Month) and Pivot Field Values (Sales).
2. Sheet 1 (Grouped Bar Chart): drag Product Name to Columns, then Month, and Sales to Rows (SUM). Drag Month to Color. Title the sheet "Quarterly Sales by Product".
3. Sheet 2 (Stacked Area Chart): drag Month to Columns and Sales to Rows (SUM). Choose the Area mark type and drag Product Name to Color to stack the products. Title the sheet "Overall Sales Trend".
4. Sheet 3 (Table): drag Product Name to Rows, Month to Columns, and Sales to Text. Use Analysis > Totals > Show Row Grand Totals to add the quarterly total.
5. Create a new Dashboard and drag all three sheets onto it, arranging the two charts at the top and the table at the bottom.
6. For interactivity, select "Use as Filter" on the charts and add a Product Name filter (Show Filter) so every view updates together.
7. Title the dashboard "Product Sales Dashboard" and save the workbook.

Insert Tableau dashboard screenshot here (Output 4)

Output 4: Product Sales Tableau Dashboard

Observation: The dashboard combines the product comparison, the overall monthly trend, and the detailed table. Selecting a product or a month filters all three views at once.

Set 6 Procedure

1. Create a matrix of monthly sales with months as rows and products as columns.
2. Use `barplot()` with `beside = TRUE` to draw the grouped bar chart.
3. Convert the data into long format and use `ggplot()` with `geom_area()` to draw the stacked area chart.
4. Create a data frame with `data.frame()`, add a total column with `rowSums()`, and display it with `print()`.
5. Connect the dataset in Tableau, pivot the month columns, build the three sheets, and combine them in a dashboard.

Set 6 Result

The monthly product sales data was successfully visualized using a grouped bar chart, stacked area chart, and data table in R. A Tableau dashboard combining all three views was created for interactive exploration.

Key Functions Used

Function / Tool	Purpose
<code>matrix()</code>	Builds the month-by-product sales table.
<code>barplot()</code>	Creates the grouped bar chart (<code>beside = TRUE</code>).
<code>ggplot()</code> + <code>geom_area()</code>	Creates the stacked area chart.
<code>data.frame()</code>	Builds the sales data table.
<code>rowSums()</code>	Calculates the total sales for each product.
Tableau Dashboard	Combines sheets with filters for interactive exploration.